

M13-C3 — Signposted future probes (H1–H4)

H1–H3 designs approved; H4 proposed / infra-blocked
(not tested / not claims)

1 Status

Near-term probes for the M13-C3 target-wise investigation
(`m13_bucket3_binary_collider_and`).

- H1 / H2: **approved** / not yet implemented / not tested / not claims
- H3: **design approved** / not yet implemented / not tested / not a Bucket 3 claim (fixture YAML not yet built)
- H4: **proposed** / blocked on target-wise cautious support / not tested / not a Bucket 3 claim
- **Not** approved Bucket 3 claims
- **Not** locked Bucket 1/2 claim content
- **Not** report-facing prose (`docs/report/findings/` is out of scope)
- Markdown twin: `future_probes.md`

Samuel decision (2026-08-01) — H1 / H2.

1. **H1 approved** — nested-prefix / missing-(0,0,0) AAMAS ablation on this fixture (e.g. $n = 25$, seed 42, targets a and b) against the approved $n = 30$ baseline.
2. **H2 design approved** — separate schema-2 fixture with isolated $D \sim \text{Bernoulli}(1/2)$ (variables a, b, c, d).
3. **H2c in scope** for the first H2 write-up.

Samuel decision (2026-08-01) — H3.

1. Graph/naming sketch **approved**.
2. Root parameters **approved**: $B \sim \text{Bernoulli}(4/5)$ ($P(B=1) = 4/5$, $P(B=0) = 1/5$); $D \sim \text{Bernoulli}(7/10)$ ($P(D=1) = 7/10$, $P(D=0) = 3/10$); $A \sim \text{Bernoulli}(1/2)$ as previously sketched.
3. Fixture id **locked**: `m13_bucket3_binary_bd_and_lead_a` (AND of parents `b,d`; BK-leading isolated `a`). Distinct from `m13_bucket3_binary_collider_and` and from the intended H2 id `m13_bucket3_binary_collider_and_iso_d` (when H2 is authored).
4. **AAMAS contrast in scope** for the H3 write-up (secondary to primary ECAI inspection of target c ; valuable as a related but distinct check from H2).

H4. Signposted only; **not** approved for implementation and **blocked** until target-wise cautious learning is supported (see H4 section).

No probe in this file has been implemented or run under this record. Approvals authorise investigation; they do not create a Bucket 3 claim or a run matrix. Planned first frozen sample

for H3 (to confirm at build): `n30_seed42`.

2 Motivation (verified facts from the AAMAS arm)

On the approved frozen sample `n30_seed42`, configuration `aamas2025` (`greedy / folding_selection(mgr) / folding_space(bk) / brave / check_ic`):

Target	Outcome	Delta (if any)
<code>c</code>	<code>solved</code>	<code>c(A) :- a_val_1(A), b_val_1(A).</code> (assumption-free)
<code>a</code>	<code>completed_no_solution</code>	<code>empty</code>
<code>b</code>	<code>completed_no_solution</code>	<code>empty</code>

Sources:

- collection summary: `causal/outputs/aba_learning/targetwise/m13_bucket3_binary_collider_and/aamas2025/n30_seed42/summary.md`
- target `c` delta: `.../cells/target-c/output/delta.aba`
- target `a/b` reports and `prolog.stdout` under `.../cells/target-{a,b}/`

For target `c`, the learned delta string coincides with the evaluator-only positive-state reference in `causal/outputs/causal_fixtures/m13_bucket3_binary_collider_and/mechanism_reference.json`. Record that as **syntactic coincidence**, not as a causal-recovery verdict.

Procedural reading of the AAMAS root-target path (verified in `.../cells/target-a/output/prolog.stdout`; target `b` is symmetric):

1. Greedy folding builds **maximal co-occurrence bodies** from BK for each E^+ row (not a minimal-parent search).
2. The both-1 co-occurrence body `a(A) :- b_val_1(A), c_val_1(A)` is accepted under brave entailment.
3. The both-0 co-occurrence body `a(A) :- b_val_0(A), c_val_0(A)` over-generalises; assumption/contrary repair is attempted.
4. Contrary rote learning cites rows **26** and **30** — the only sample atoms (0,0,0) in `samples/n30_seed42.csv` — and contrary folding then fails, yielding `completed_no_solution`.

These facts motivate probe families H1 and H2. They do not by themselves approve a Bucket 3 claim.

2.1 Motivation addendum (ECAI arm — for H3)

On the same frozen sample `n30_seed42`, configuration `ecai2024` (`folding_mode(nd)`, `folding_selection(any)`, `brave / check_ic`), target `c` **solved** with delta (verified):

```
c(A) :- alpha_1(A), a_val_1(A).
c_alpha_1(A) :- b_val_0(A).
assumption(alpha_1(A)).
contrary(alpha_1(A), c_alpha_1(A)) :- assumption(alpha_1(A)).
```

In `.../ecai2024/n30_seed42/cells/target-c/output/prolog.stdout`, the first `nd` fold on an early E^+ row replaces the sample id by `a_val_1(A)` (the first BK predictor under declared order

a, b for that cell), then assumption repair retains a in the ordinary target body while the second parent enters via a contrary. Collection summary: `.../ecai2024/n30_seed42/summary.md`.

This motivates **H3**: put an *independent* distractor first in BK by naming, under ECAI, on a new fixture. H3 design is now approved (see Status); it is still **not** a Bucket 3 claim and is **not yet built**.

3 Ordering

1. **H1 first** — nested-prefix / missing-(0,0,0) ablation on the existing fixture and AAMAS config (seed 42). **Investigation approved**.
2. **H2 second** — separate schema-2 fixture with isolated covariate D on the $A \rightarrow C \leftarrow B$ AND collider. **Design approved**; intended id `m13_bucket3_binary_collider_and_iso_d` (author when H2 is built); sample n /seed still to lock at H2 authorship.
3. **H3** — separate schema-2 fixture `m13_bucket3_binary_bd_and_lead_a` with BK-leading independent A and AND collider $B \rightarrow C \leftarrow D$. **Design approved**; primary ECAI target c ; **AAMAS contrast in scope**. YAML / sample / learning not yet produced.
4. **H4 (proposed)** — same fixture/`n30_seed42`; ECAI **root** targets under **cautious** vs locked brave; **blocked** on target-wise cautious support. Not approved for run until infra is designed and approved.

Do not expand any family into a broader run matrix beyond these approved probes (and a later-approved H4) without a further Samuel decision.

3.1 Relation among probe families (keep distinct)

Probe	Focus
H1	AAMAS roots; missing negative witness / both-0 over-generalisation on the current three-variable fixture
H2	AAMAS target c ; irrelevant covariate enters greedy maximal co-occurrence bodies (minimality vs solvability); intended id <code>m13_bucket3_binary_collider_and_iso_d</code>
H3	ECAI target c (primary); BK order puts independent distractor first ; nd first-fold / repair path; AAMAS contrast in scope ; id <code>m13_bucket3_binary_bd_and_lead_a</code>
H4	Same fixture/sample ; ECAI root targets a/b ; brave $\rightarrow$ cautious ; requires target-wise learning-mode support

H3 is not a duplicate of H2: different DAG, different distractor placement/name, different primary config (ECAI nd vs AAMAS greedy), and distinct fixture ids. The in-scope AAMAS arm on H3 still differs from H2 because the irrelevant variable is BK-leading a and the mechanism parents are b, d .

H4 is not a duplicate of H1 or H3: it holds the table fixed and varies **entailment** / **learning mode** on the target-wise path for **roots**.

4 Probe family H1 — support sensitivity on AAMAS root targets

Status: **approved** / not yet implemented / not tested / not a claim.

4.1 H1a — spurious root rule under missing negative witness

On this fixture, AAMAS, target a (symmetrically b), seed 42: if the table is a nested prefix that **omits all** $(0,0,0)$ **rows** (e.g. $n = 25$), learning is predicted to **solve** with a delta that includes the both-0 co-occurrence rule

$a(A) :- b_val_0(A), c_val_0(A).$

(respectively $b(A) :- a_val_0(A), c_val_0(A).$), which is **false** as a population implication and is blocked on the approved $n = 30$ table by E^- rows 26 and 30.

Nested-prefix fact on the approved sample (seed 42): rows 1–25 contain no $(0,0,0)$ atom; rows 26 and 30 are the only $(0,0,0)$ observations in $n = 30$.

4.2 H1b — negative witness in the unsafe cell

If the sample contains at least one opposite-label row in that both-0 predictor cell, the both-0 rule fails brave entailment under the observed AAMAS path. On approved $n = 30$ this manifested as **completed_no_solution**, not as emission of only the safe both-1 rule.

4.3 Explicit non-claims for H1

- Full support does **not** imply AAMAS will emit a “correct” deterministic root mechanism rule: roots have **no_observed_parent_deterministic_rule** in the evaluator reference.
- “One of each population atom” is a sufficient practical condition on this approved sample, but the operational point is a **label conflict inside the greedy candidate body cell**, not a generic demand for full joint support.
- H1 does not claim that omitting $(0,0,0)$ is the only way AAMAS can emit a spurious root rule, nor that every seed behaves identically.

4.4 Planned investigation (H1)

Approved. Nested-prefix / missing- $(0,0,0)$ ablation on the existing fixture **m13_bucket3_binary_collider_and** and AAMAS config, seed 42 (illustrative first cut: $n = 25$), comparing root-target outcomes and deltas to the approved $n = 30$ baseline. No broader matrix beyond this probe without further approval.

5 Probe family H2 — irrelevant isolated covariate on AAMAS target c

Status: **design approved** / not yet implemented / not tested / not a claim. Requires a **new** schema-2 fixture before any learning run.

5.1 H2a — irrelevant covariate enters greedy bodies

Extend the AND collider with an isolated binary root $D \sim \text{Bernoulli}(1/2)$, independent of $\{A, B, C\}$. Under AAMAS, when both values of D appear among E^+ for c , greedy catalogue bodies are predicted to **include a d_val_* literal**; the learner is **not** expected to return the D -free evaluator-form rule

$c(A) :- a_val_1(A), b_val_1(A).$

even if an equivalent covering theory exists.

5.2 H2b — learnability vs minimality

On that extension, AAMAS target c **may still solve** (e.g. two D -conditioned rules covering $d = 0$ and $d = 1$ with $a = b = 1$, or an assumption repair). The failure mode under test is **failure to omit an irrelevant predictor / recover the minimal AND form**, not necessarily `completed_no_solution`.

Do **not** record the strong claim “ c becomes unlearnable” as the primary hypothesis. Treat strong “unlearnable” wording as a probe that may be **refuted**.

5.3 H2c — sample-edge (in scope)

If E^+ for c realise only one value of D , AAMAS may emit a single spuriously D -conditioned conjunction that is sample-adequate but population-inadequate.

Decision: H2c is **in scope** for the first H2 write-up (not deferred).

5.4 Planned investigation (H2)

Design approved (2026-08-01). Remaining implementation steps (not performed by this documentation update):

1. Lock remaining fixture parameters as needed (sample n and seed; intended id `m13_bucket3_binary_collider_and_iso_d`) and author a separate schema-2 YAML with variables `a,b,c,d` and isolated $D \sim \text{Bernoulli}(1/2)$ on $A \rightarrow C \leftarrow B$. Do **not** collide with H3 id `m13_bucket3_binary_bd_and_lead_a`.
2. Certify population / mechanism reference as for M13-C3.
3. Run AAMAS target-wise on c (and optionally roots) to test **H2a/H2b**, and include **H2c** in the first H2 write-up (construct or select a sample edge if it does not arise naturally).
4. No portfolio or multi-fixture matrix until that single extension is inspected.

6 Probe family H3 — ECAI first-fold distraction by BK-leading independent variable

Status: **design approved** / not yet implemented / not tested / not a Bucket 3 claim. Fixture YAML **not yet built**. Do not reuse or alter `m13_bucket3_binary_collider_and`.

6.1 Approved fixture design

Field	Value
Fixture id	m13_bucket3_binary_bd_and_lead_a
Internal / display	a,b,c,d / A, B, C, D
DAG	$B \rightarrow C \leftarrow D$; no edges involving A
A	Bernoulli(1/2); independent of $\{B, C, D\}$
B	Bernoulli(4/5) — $P(B=0) = 1/5$, $P(B=1) = 4/5$
D	Bernoulli(7/10) — $P(D=0) = 3/10$, $P(D=1) = 7/10$
C	deterministic $C = B \wedge D$
Evaluator reference for c	$c(A) :- b_val_1(A), d_val_1(A)$. (a must not appear)
BK order for target c	predictors a , then b , then d (irrelevant first)
Planned first sample	n30_seed42 (confirm at build)
Primary learning arm	ECAI target c
Secondary arm	AAMAS on the same frozen sample (in scope)

Naming is part of the probe: under exact-value encoding, a precedes the mechanism parents in BK.

6.2 Scientific question

When learning target c under **ECAI** (`folding_mode(nd)`, one-step folds, `folding_selection(any)`, etc.) on a frozen sample from this fixture, does BK-leading independent a **distract** the first fold (and/or the final ABA shape) away from the mechanism-aligned B, D conjunction?

Secondary (in scope): on the **same** sample under **AAMAS**, does greedy maximal co-occurrence include `a_val_*` in bodies for target c , and how does that compare to the ECAI path and to H2 (where the distractor is trailing d on parents a, b)?

Motivated by the completed ECAI c cell on `m13_bucket3_binary_collider_and` (see Motivation addendum above).

6.3 Working hypothesis (signpost only; not a claim)

Under ECAI on target c , the first `nd` fold is predicted to match an `a_val_*` literal from an early E^+ row because a precedes b and d in BK. That one-literal body is not a valid covering rule for $C = B \wedge D$. Subsequent assumption/contrary repair may still solve, but the learned framework is predicted to **retain a in the ABA presentation** and **not** coincide with the evaluator string `c :- b_val_1, d_val_1`, unless later steps drop a (to be observed, not assumed).

Under AAMAS on the same sample, greedy bodies for E^+ rows with $C = 1$ (hence $B = D = 1$) are predicted to cite the co-occurring value of a as well, so the A -free evaluator form is not expected as the emitted delta (learnability vs minimality / distractor inclusion — related to but distinct from H2).

6.4 Explicit non-claims for H3

- H3 does not assert that ECAI or AAMAS cannot solve, nor that a can never be dropped later in the transformation sequence.

- H3 does not reuse H2’s intended graph ($A \rightarrow C \leftarrow B$ with isolated D); the distractor here is A , and parents of C are B, D .
- Completing H3 would still not by itself approve a Bucket 3 claim.

6.5 Planned investigation (H3)

Design approved (2026-08-01). Remaining implementation steps (not performed by this documentation update):

1. Author schema-2 YAML `m13_bucket3_binary_bd_and_lead_a` with the parameters above; planned sample `n30_seed42`.
2. Build population/certificate/mechanism reference; confirm evaluator reference for c omits a .
3. Run **ECAI** target-wise on the frozen sample with primary inspection of **target** c (BK order, first fold in `prolog.stdout`, `delta.aba` vs evaluator reference).
4. Run **AAMAS** on the same frozen sample as an **in-scope** secondary contrast (distractor inclusion under greedy maximal bodies; distinct from H2’s trailing- d design).
5. No portfolio or multi-fixture matrix beyond this approved H3 package without further approval.

7 Probe family H4 — cautious vs brave on ECAI root targets

Status: **proposed** / blocked on target-wise cautious support / not tested / not a Bucket 3 claim.

Uses the **existing** fixture and frozen sample `m13_bucket3_binary_collider_and` / `n30_seed42`. Does **not** require a new DAG. Cannot be executed with today’s target-wise validator as-is.

7.1 Motivation (verified facts)

On ECAI **brave** (`configs/ecai2024_config.pl: learning_mode(brave)`), targets a and b **solved** with assumption-rich deltas (summary: 11 delta rules, 3 assumptions each). Artefacts:

- `.../ecai2024/n30_seed42/cells/target-a/`
Files: `delta.aba`, `prolog.stdout`, and `bk.sol_chk.asp`
- mirror: `.../cells/target-b/`
- collection summary: `.../ecai2024/n30_seed42/summary.md`

Verified sample atoms (seed 42):

Row	(A, B, C)	Role for target a
9	$(1, 0, 0)$	E^+
26	$(0, 0, 0)$	E^-

Rows 9 and 26 share identical BK literals `b_val_0`, `c_val_0`. No deterministic rule over (B, C) assigns both labels. The brave delta for a includes the α_2/α_3 nest (verified in `delta.aba`):

```
a(A) :- alpha_2(A), b_val_0(A).
c_alpha_2(A) :- alpha_3(A), c_val_0(A).
c_alpha_3(A) :- alpha_2(A), b_val_0(A).
```

Trace/fixture integration reading (for the signpost): this is a **per-row assumption choice gadget** — one stable model can Hold at 9 and Reject at 26 because ground assumption atoms

differ. Runner `solved` means brave coverage of E^+/E^- , **not** recovery of a functional root mechanism (`no_observed_parent_deterministic_rule`).

Engine fact (`asp_engine.pl`, `entails/5`): brave entailment requires existence of a stable model under joint E^+/E^- constraints; cautious entailment requires E^+ atoms to lie in **cautious consequences** (true in every answer set) and E^- atoms not to. Under that reading, the current brave delta should **fail** cautious entailment: e.g. $a(9)$ is not true in every stable model of the choice gadget. Cautious re-check after contrary rote is invoked in `gen.pl` (`gen6`).

7.2 Infrastructure prerequisite (required before H4 can run; not yet done)

H4 is **not** “point YAML at a cautious `.pl` and run.” The **target-wise** path currently **hard-requires brave**.

Verified code fact (`causal/targetwise/config.py`, `_read_learning_mode`): the runner reads `set_lopt(learning_mode(...))` from the Prolog config and **rejects** any mode other than **brave**, with an error that the current target-wise runner supports brave learning only. The target-wise README states that the supported regime selects brave learning. Existing configs only supply `learner.prolog_config` (e.g. `configs/ecai2024_config.pl`); learning mode is **not** a free YAML field — it is derived from the consulted `.pl` file and then validated.

Two-stage plan:

1. Infrastructure / config design (prerequisite, careful, small)

Decide how cautious mode is exposed for target-wise without breaking existing brave collections or hashes. Likely pieces (for the record; final design needs Samuel/implementation approval):

- a **new** Prolog options file (do **not** edit `configs/ecai2024_config.pl` in place), e.g. an ECAI-cautious twin that keeps `nd / any / all / relto / check_ic` but sets `learning_mode(cautious)`;
- relax or extend `causal/targetwise/config.py` (and README/tests) so `learning_mode(cautious)` is an **explicitly allowed** target-wise mode when selected that way;
- a **new** `configuration.id` and output directory so brave `ecai2024` artefacts remain untouched;
- confirm what `check_ic / .sol_chk.asp` / the post-run artefact audit mean under cautious learning (joint existential SAT vs cautious consequences) so diagnostics are not misread.

Mode is coupled to Prolog consult, config hashing, manifests, and the brave-only validator. Do not imply that only a YAML one-liner is required.

2. Scientific run (only after the above)

Target-wise collection on the existing fixture/`n30_seed42` for roots a/b (optional c control) under the new cautious configuration.

7.3 Precise hypotheses (not claims)

H4a (current delta is brave-specific). The serialized ECAI brave delta for target a on this sample does **not** cautiously entail the E^+/E^- set: in particular, both-0 E^+ such as $a(9)$ are not cautious consequences of that framework.

H4b (cautious learning changes the root outcome). Once target-wise cautious mode is supported, re-running target a (and symmetrically b) on the **same** frozen sample with ECAI-like folding options but `learning_mode(cautious)` yields an outcome **different** from the brave solve: either `completed_no_solution`, or a solved delta **qualitatively different** from the 11-clause

α_2/α_3 choice gadget (e.g. more rote residual structure on the ambiguous (*other*, *c*) = (0, 0) cell). The brave choice-gadget solution is **not** expected to reappear unchanged.

H4c (fixture link — non-claim). Interpret any difference against the AND-collider fact that roots have **no** deterministic mechanism over the other observed variables, and that $(B, C) = (0, 0)$ (resp. $(A, C) = (0, 0)$) is label-ambiguous for the root target. H4 tests **learning-mode** / **entailment semantics** on the target-wise path, not DAG recovery.

7.4 Explicit non-claims for H4

- Not claimed that cautious mode is “better” for causal recovery.
- Not claimed that cautious mode will recover a correct root mechanism rule (none exists).
- Not a ranking of brave vs cautious as general ABALearn policy beyond this controlled probe.
- Not claimed that the probe can be executed with today’s target-wise validator as-is.

7.5 Planned investigation (when H4 is started)

Not approved now. When Samuel opens H4, proceed in two stages after resolving the deferred decisions below:

1. **Infra:** implement cautious support for target-wise (new `.pl` + validator/README/tests + distinct `configuration.id`); leave brave `ecai2024` outputs immutable.
2. Optionally cheap **H4a** pre-check: cautious-consequence inspection of the existing brave framework without a full re-learn.
3. **H4b run:** target-wise cautious collection on `n30_seed42` for *a/b* (optional *c*).
4. Compare outcomes, deltas, and trace entailment KO points to the locked brave ECAI cells.

7.6 Decisions deferred until H4 begins (flagged; not decided now)

These must be resolved **when H4 is started**, not in the current H1–H3 work. They are recorded here so the infra prerequisite is not forgotten:

1. **Cautious Prolog filename** — e.g. a twin such as `configs/ecai2024_cautious_config.pl` (illustrative only); do **not** edit `configs/ecai2024_config.pl` in place.
2. **New target-wise configuration.id** and output path so brave `ecai2024` collections and hashes remain immutable.
3. **Validator change** — approve relaxing `causal/targetwise/config.py` (`_read_learning_mode`) so `learning_mode(cautious)` is explicitly allowed when selected via that `.pl`; update README/tests accordingly.
4. `check_ic` / `.sol_chk.asp` / **post-run artefact audit** under cautious learning — how to interpret joint existential SAT vs cautious consequences so diagnostics are not misread.
5. **H4a pre-check in scope?** — optional cautious-consequence inspection of the locked brave framework before a full re-learn.
6. **Optional target *c* control** on the cautious collection, or roots *a/b* only.

Until those are decided and infra lands, H4 remains **proposed** / **infra-blocked** / **not tested** / **not a claim**.

8 Boundary distinctions to preserve

Keep separate throughout any future probe write-up:

Object	Role
AAMAS strategy / greedy maximal co-occurrence search	procedural learner behaviour (H1/H2)
ECAI nd first-fold / BK serialisation order	procedural learner behaviour (H3)
Brave vs cautious entailment / learning mode	procedural / semantic learner behaviour (H4)
Finite-sample support and label conflicts in candidate body cells	sample information
Mechanism correspondence for c (AND)	evaluator-only interpretation of a non-root
Absence of deterministic root mechanisms	evaluator reference status
Syntactic delta = reference string	coincidence, not automatic causal verdict
Runner solved under brave	brave coverage of $E^\pm$, not mechanism recovery

Do not treat predictive rules for roots as mechanism recovery.

9 Cross-links

- Investigation record: `experiment.md`
- Mathematical fixture dossier: `fixture_dossier.tex`
- Bucket 3 planning hub: `../M1.3-bucket3-claims.md` (still **no claim**)
- Research decision log: `docs/research/decisions.md` (2026-08-01 H1–H3 entries)
- Fixture pre-run bundle: `causal/outputs/causal_fixtures/m13_bucket3_binary_collider_and/`
- AAMAS baseline: `causal/outputs/aba_learning/targetwise/m13_bucket3_binary_collider_and/aamas2025/n30_seed42/`
- ECAI baseline (H3/H4 motivation): `causal/outputs/aba_learning/targetwise/m13_bucket3_binary_collider_and/ecai2024/n30_seed42/`
- Target-wise brave-only validator: `causal/targetwise/config.py`
- Entailment implementation: `asp_engine.pl` (entails/5)

10 Decisions recorded

Item	Decision	Date
H1 nested-prefix / missing-(0, 0, 0) AAMAS ablation	Approved	2026-08-01
H2 four-variable isolated- D fixture design	Approved	2026-08-01
H2c in first H2 write-up	In scope	2026-08-01
H3 graph/naming; B 80/20, D 70/30; id <code>m13_bucket3_binary_bd_and_lead_a</code> ; AAMAS contrast	Design approved	2026-08-01
H4 cautious vs brave on ECAI roots	Proposed only (blocked on infra)	—

11 Remaining before implementation

- **H1:** execute the approved ablation and record outcomes.
- **H2:** author `m13_bucket3_binary_collider_and_iso_d` (or confirm that id); lock sample (n, seed); certify; run AAMAS including H2c.
- **H3:** author `m13_bucket3_binary_bd_and_lead_a`; build planned `n30_seed42`; certify; run **ECAI** (primary) and **AAMAS** (in-scope contrast) with inspection of target c .
- **H4:** deferred until H4 is opened. Infra/config decisions are **flagged** in the H4 section (“Decisions deferred until H4 begins”) and are **not** to be resolved during current H1–H3 work. No cautious collection until then.
- Still **no** Bucket 3 claim and **no** expanded run matrix without a further Samuel decision.